PARTH KULKARNI

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SUMMARY

AI Engineering student experienced in building and evaluating LLM inference and RAG pipelines, deploying open-weight models, and working with embeddings, vector databases, GPUs, containers, and cloud infrastructure — with strong Python fundamentals and a track record of debugging model failures and documenting experiments for reproducibility.

EDUCATION

JSPM's Rajarshi Shahu College of Engineering (RSCOE), Pune **Expected 2028**

B.Tech in Computer Engineering | CGPA: 8.11/10

Relevant Coursework: Data Structures & Algorithms, DBMS, OOP, Distributed System Design, Linear Algebra & Probability

Mahatma Phule Junior College, Ahmadpur **2024**

HSC (Class XII) | MHT-CET Percentile: 95.25

PROJECTS

ToolBot — LangGraph + MCP Multi-Agent AI Chatbot **2026**

Python, LangChain, LangGraph, MCP, FastAPI, REST APIs, pytest, LLM Evaluation, Git/GitHub

- Built a **multi-agent chatbot with LangGraph** where the LLM autonomously selects tools across **hosted MCP servers (Context7, DeepWiki, Semgrep)** and a custom **FastAPI arithmetic tool server**, replacing hard-coded tool logic with dynamic, context-aware task execution.
- Engineered **prompt and tool-routing logic** to fix multi-tool chaining failures and **cut hallucinated tool calls**; restructured the codebase into modular **tools/, prompts/ and evals/** layers behind a **FastAPI API layer**.
- Implemented **automated testing (pytest)** and an **evaluation suite** to validate tool selection and response relevance, catching regressions before changes; documented architecture, setup and results in a README for code review.

KisanSetu — Agentic Multimodal AI Platform for Crop Disease Diagnosis **2026**

Python, FastAPI, LangChain, Gemini 2.5 Flash, Vector Databases, ResNet50, Google Cloud Platform, Twilio

- Solved **hallucinated crop-disease advice from LLMs** by building a **RAG pipeline** (embeddings + vector database, LangChain) over verified knowledge bases to ground **Gemini 2.5 Flash** responses, served through **FastAPI REST APIs**.
- Added an **offline ResNet50 image-classification fallback** so farmers in **low-connectivity areas** still get a diagnosis; benchmarked quality vs latency against the cloud multimodal (image + text) path.
- Deployed **containerised on Google Cloud Platform** with a **multilingual Twilio IVR** so non-smartphone, regional-language users can reach diagnosis by phone.

ResumeAgent — AI Resume Analyser, ATS Ranker & Skill-Gap Detector **2026**

Python, LangGraph, FastAPI, React, Vite, Tailwind CSS, Pydantic, Supabase, Docker, Render

- Solved the **opaque ATS-rejection problem**: built an **LLM-powered analyser** (FastAPI + React) that parses **PDF/DOCX** resumes, scores them against a job description and shows a **visual skill-gap matrix**.
- Designed a **hallucination detector** that flags claimed skills lacking project proof, an **agentic LangGraph workflow**, and a **Gemini-primary, Groq-fallback LLM layer** for reliability.
- Added **career roadmaps**, a **live A4 resume editor** with AI enhancement and **PDF export**, and **Supabase**-backed history; shipped with **Docker on Render**.
- Co-authored the research paper “AI-Powered Career Diagnostics and Resume Enhancement System” (under review).

TECHNICAL SKILLS

Languages: Python, C++, C, JavaScript, TypeScript, SQL

AI Engineering: LLMs, LangChain, LangGraph, RAG Pipelines, AI Agents, Multi-Agent Systems, Model Context Protocol (MCP), Prompt Engineering, LLM Evaluation & Observability, Model Inference & Serving, Embeddings, Open-Weight Models, Regex-based Information Extraction, NLP

Software Engineering: REST API Design (FastAPI, Flask), Distributed System Design, React, HTML/CSS, Git/GitHub, Docker, Containers, CI/CD (GitHub Actions), Debugging & Testing, Technical Documentation

Databases & Cloud: MySQL, Vector Databases, Supabase, IndexedDB, Google Cloud Platform, Cloud Deployment & Infrastructure, GPUs

Machine Learning: Machine Learning, Deep Learning, Scikit-learn, TensorFlow, Pandas, NumPy, Model Evaluation & Benchmarking

ACHIEVEMENTS & ACTIVITIES

- Competed in 5+ national/college hackathons (incl. DevClash 2026, IBM Dev Day: AI Demystified) delivering functional, production-oriented LLM-powered prototypes as part of small cross-functional engineering teams under 24–48 hour constraints.
- Published 10+ datasets and 20+ notebooks on Kaggle covering supervised learning, EDA, model evaluation, and feature engineering, including one dataset with 20k+ views and 4k+ downloads (Core Member, ML & Data Science Team, Coding Club RSCOE).

CERTIFICATIONS

Oracle Certified Foundations Associate – Agentic AI (Oracle)

Machine Learning with Python (IBM)

Oracle Cloud Infrastructure 2025 Certified Data Science Professional